

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Urogenital Protozoal Flagellates *Trichomonas Vaginalis*

First Course Lecture 6

Trichomonas vaginalis

Objective

- **Disease:** Trichomoniasis (STD)
- **The causative agent and diseases:** the most important are three species of *trichomonas*:
 1. *Trichomonas vaginalis* cause (STD) transmitted by sexual intercourse, has no cyst stage, habitat in urogenital system
 2. *Trichomonas hominis* cause intestinal Trichomoniasis, transmitted orally, has cyst stage, habitat in human intestine
 3. *Trichomonas tenax* causes gingivitis, transmitted by contaminated buccal (mouth) cavity, has no cyst stage, habitat in oral (mouth) cavity
- **Infective stage of vaginal Trichomoniasis :** Trophozoite stage.
- **Host:** only human body.
- **Diagnostic stage of vaginal Trichomoniasis :** Trophozoite

Introduction

Trichomonas vaginalis was first observed by Alfred Donne in 1836 in vaginal secretion, it is sexual transmitted diseases (STD), and sexual transmission is the usual mode of infection. Humans are the only known host, without cyst form and does not survive well in the external environment. There are more than 100 species of *Trichomonas* only three species infect human beings, most important are three species of *Trichomonas*:

- ***Trichomonas vaginalis*** causes **vaginal trichomoniasis** and habitat in human vagina and urethra (urogenital system).
- ***Trichomonas hominis*** causes **intestinal trichomoniasis** and habitat in human intestine.
- ***Trichomonas tenax*** causes **gingivitis trichomoniasis** and habitat in oral cavity in gum, between teeth and calcified matter.



It is flagellates protozoa without kinetoplast like (***Giardia lamblia***), infective stage are trophozoite and no cyst stage.

Mode of transmission is by person (patient) to person by sexual intercourse. Babies may get infected during birth and cause conjunctivitis.

Transmission often coexists with other sexual transmission diseases like vaginal candidiasis, Gonorrhea (GC), syphilis and human immunodeficiency virus (HIV), Incubation period takes about 4 days to 4 weeks. It is an obligate protozoan parasites only in urogenital system.

Morphology

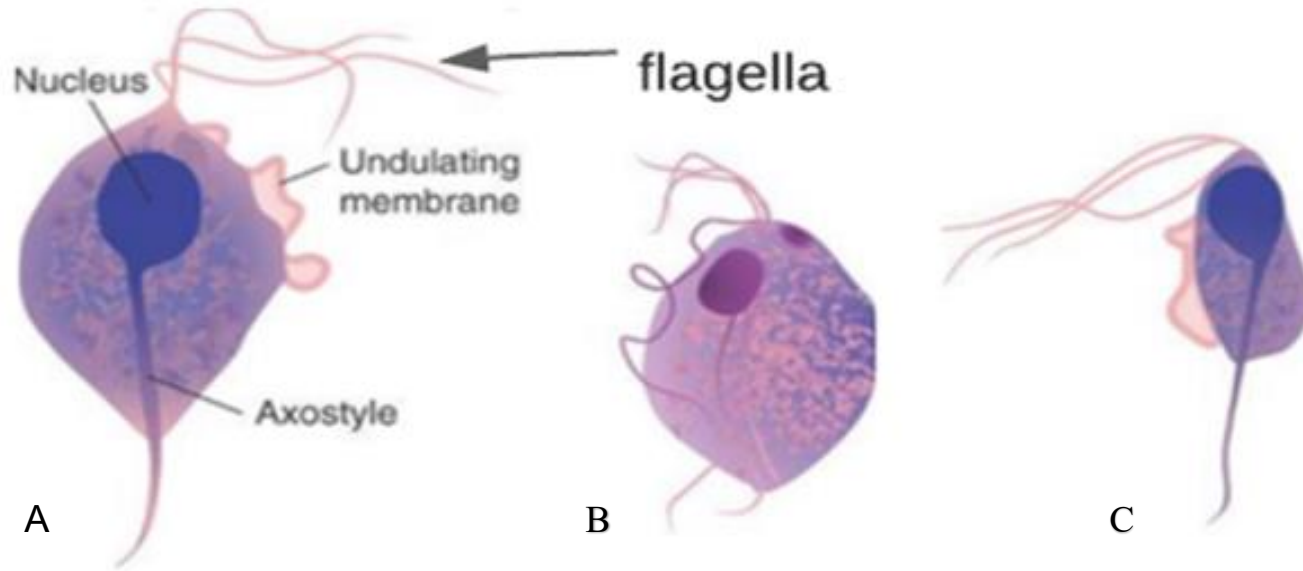
Trichomonas species has one stage and it is the trophozoite stage and considered the infective stage for humans and morphology of all species are the same.

Trophozoite is spindle or spiral, oval (ovoid) or pear shaped, measures 10-30 μm long and 5-10 μm wide, colorless and have 4 free flagellate in anterior end with one costa and posterior oxostyle and has one short undulating membrane reaching up to the middle of the body which gives rise to the spiral movement.

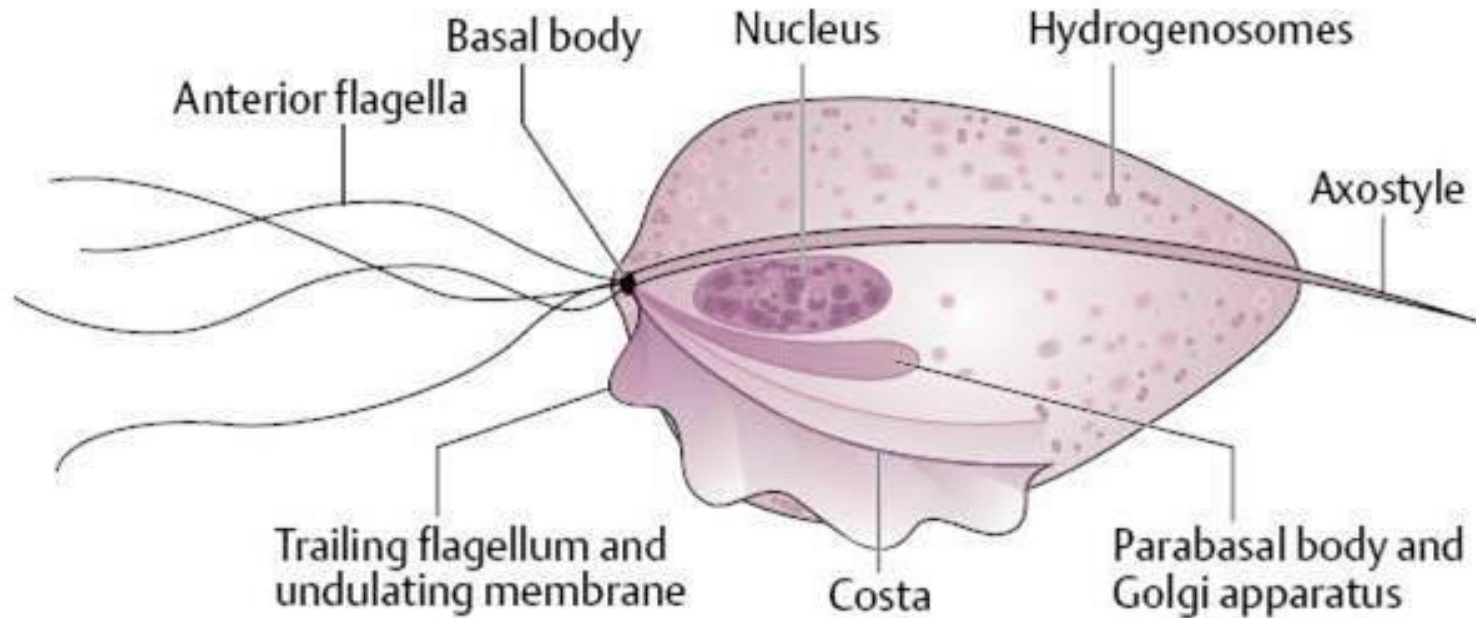


The trophozoites have granular cytoplasm include one nucleus oval or spherical in shape (according to the species) at the middle line near the anterior end. Para basal bodies are present around nucleus in the anterior part. ***Trichomonas vaginalis*** live in acidic conditions (PH 4.5) and it multiplies by longitudinal binary fission.

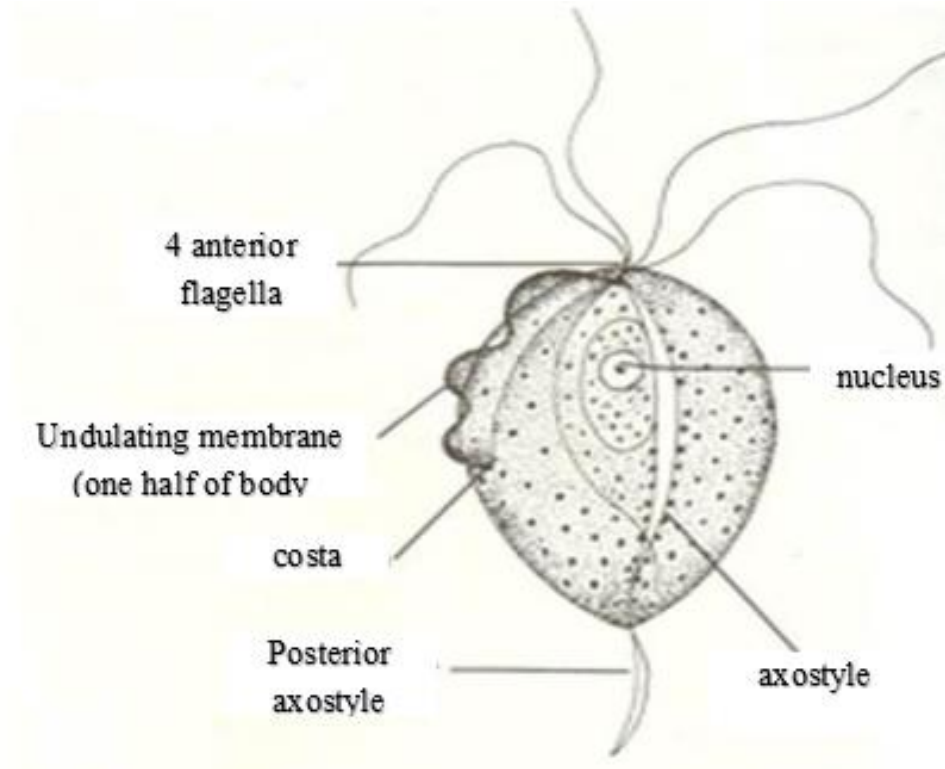
The three species can be differentiated from each other by their sites in human body, morphology shape of nucleus, size, and long of undulating membrane. It is motile with a rapid spiral movement as the falling of leaf trees. It is live in human mucous membrane of genital system (male & female) and sometime urinary system.



Morphology of *Trichomonas* species, A- *T. vaginalis*, B- *T. hominis*, and C- *T. tenax*



Morphology of *Trichomonas vaginalis* (trophozoite stage).



Trophozoite stage *Trichomonas vaginalis*

Life cycle

The life cycle of ***Trichomonas vaginalis*** is completed in one human host either male or female. The trophozoite (infective stage) cannot survive outside and infection has to be transmitted directly through sexual intercourse.

The trophozoite grow by binary fission in mucosa of vagina or urethra. *Trichomonas vaginalis* habitat in females it live in mucosa of vagina, cervix, Bartholin's glands, urethra, and urinary bladder. In male it habitat mainly in the interior urethra, but may also be found in prostate, and urinary tract.

A. *Trichomonas vaginalis*

T. vaginalis is a high incidence disease in prostitutes and sexually promiscuous groups and it is found in all climates and social groups, is considered one of the most common sexually transmitted disease (STD) around the world, transmitted during sexual intercourse or when using internal contaminated clothes mainly used by female and by contaminated water, is usually found on the surface of the squamous epithelium but not columnar epithelium of uro – genital tract and it is not invasive, and causes disease in urogenital system and no extra genital disease, is the largest one in size, about $27 \times 18 \mu\text{m}$ with oval nucleus, rounded anterior end and posterior end pointed, the undulating membrane is small and reach to half of the body.



Pathogenesis and clinical features

When trophozoites reach vaginal mucosa, it starts to grow in acidic condition and feed on cells and bacteria in mucus membrane, and reproduce by longitudinal binary fission at the upper vagina to the cervix and there is no invasion on the wall of vagina. Its particularly infect squamous epithelium and not columnar epithelium.

The trophozoite secretes cysteine, proteases, lactic acid, and acetic acid which disrupt the glycogen levels and lower the pH (acidic) of vaginal fluid. The incubation period is about 1-4 weeks after the infection which causes **acute or chronic trichomoniasis**.

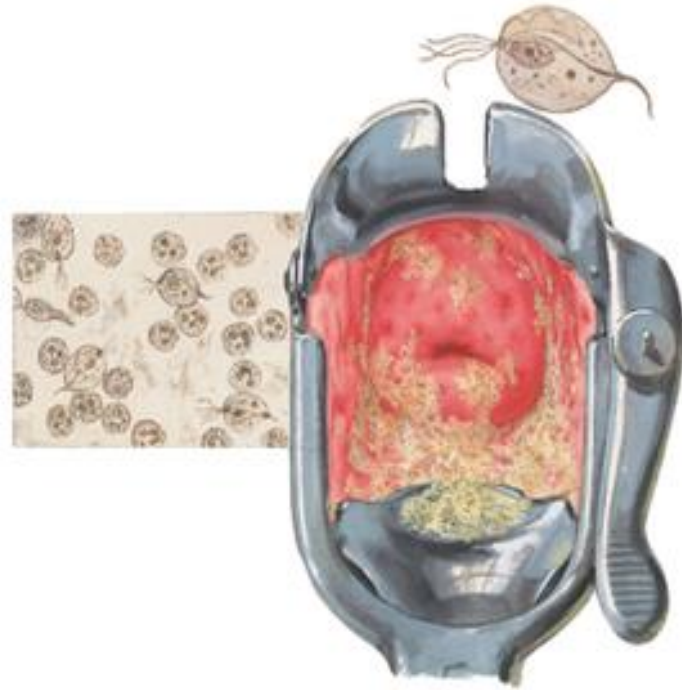
Female Trichomoniasis have heavy infection or severe inflammation of the mucosal tissue of vagina, vulva, and causes severe pruritus, with a characteristic a yellowish green often frothy discharge, an offensive bad odor, acidic (4.5 PH), itching, local irritation, pain, congestion, urethritis, cystitis and may cause dysuria. Vulva and **vaginal trichomoniasis** may lead to painful coitus (dyspareunia).

Cervical erosion is common and may be complicated by bacteria, fungi or spirocetical infection. The most important complication of trichomoniasis are endometritis and pyosalpingitis

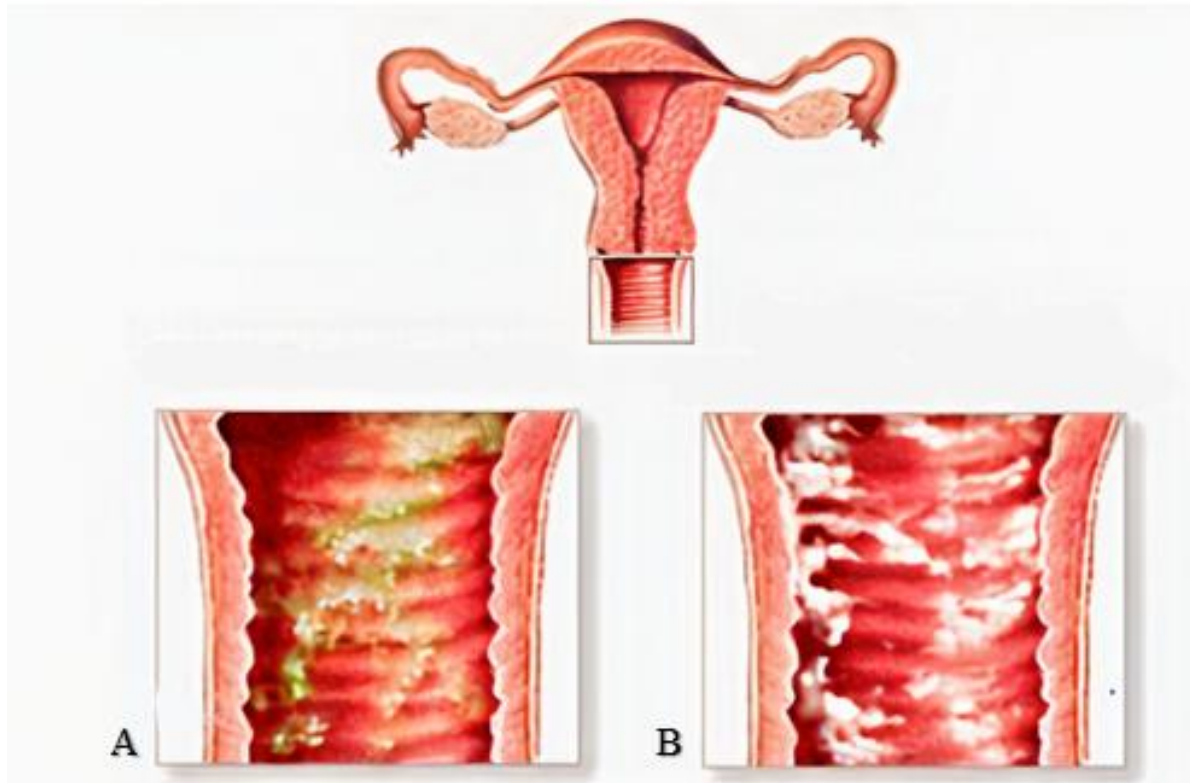
Trichomonas vaginalis causes petechial hemorrhage (strawberry-mucosa) and causes metaplastic change and desquamation of vaginal epithelium, and may be causes intracellular edema and so called chicken - like epithelium is the most characteristic feature of **trichomoniasis**.



Trichomonas vaginalis



Clinical diagnosis of vaginal trichomoniasis



The difference between A. Vaginal trichomoniasis and B. Vaginal candidiasis



Male Trichomoniasis usually have mild clinical manifestation, it can cause male urethritis and inflammation of seminal vesicles leading to burning sensation mainly during micturition and production of purulent white discharge with dysuria and causes epididymitis and prostatitis.

The most important complications of female and male trichomoniasis:

Greater chance of having an infected uterus and fallopian tubes, this infection is called pelvic inflammatory disease, which can cause infertility, cervical or prostate cancer, prostatitis, epididymitis, and may also cause bleed after intercourse.

B. *Trichomonas hominis*

Davain in 1860 and Leuckart in 1879 was the first to identifies the ***trichomonas hominis*** trophozoite, It is similar to *T. vaginalis* from morphology and life cycle but smaller in size, and contains a spherical nucleus with long undulating membrane in trophozoite, with cyst stage characterized by thick wall, both trophozoite and cyst stage are demonstrated in the semi formed stool. It is considered as non-pathogenic to human body (normal flora), if present in large numbers it may cause symptom it present spectrally in ileum and cecum, and of the important symptoms is abdominal pain and diarrhea.

C. *Trichomonas tenax*

It is also similar to ***T. vaginalis*** and ***T. hominis***, but it is the smallest one with a size of 6-10×3-4 μm. It is present normally in the mouth cavity mainly in the gum and between teeth and in the calcified matter.

It is non-pathogenic but when present in large numbers mainly when teeth are extracted it can cause congestion and redness of gum it is known to play a pathogenic role in necrotizing ulcerative gingivitis it is transmitted by kissing, and through salivary droplets by trophozoite stage (no cyst stage).



Clinical diagnosis of *Trichomonas tenax*

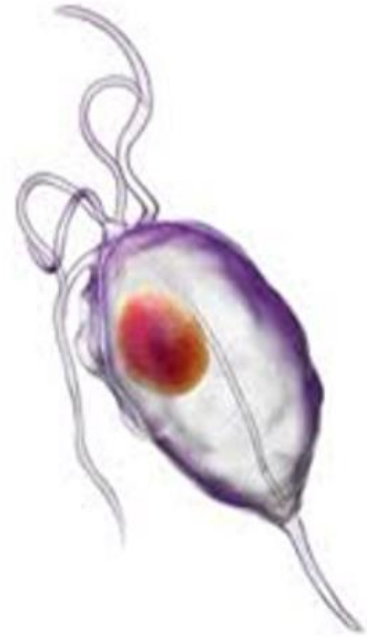


Laboratory diagnosis

Direct examination:

Microscopic examination of vaginal swab, urethral, seminal fluid, fluid from prostate massage, deposit of urine and direct examination in saline wet mount preparation, fixed smears may be stained with Giemsa stain, shows the trophozoites motile by spiral movement as the falling of leaf trees. In male the trophozoites may be found in urine or prostatic secretions.

Direct fluorescent antibody (DFA) it is another method of detection of *Trichomonas vaginalis* and is more sensitive than wet preparation.



Microscopic diagnosis of *T. vaginalis* trophozoite in vaginal swab

Indirect examination

Culture: The culture is recommended when direct microscopic examination is negative, is the most sensitive and the best method for diagnoses of *Trichomonas vaginalis* cultured in specific media, solid or liquid with antifungal and anti-bacterial like cysteine peptone liver, and maltose media (CPLM), best grow at 35-37 C° under anaerobic condition the optimal pH for growth is 5.5 - 6.0 PH for 3-5 days and less than one week.

Molecular method: by DNA hybridization and (PCR) it is highly sensitive and more specific test for diagnosis of *Trichomonas vaginalis*.

Serological test: by Elisa is used for demonstration of *Trichomonas vaginalis* antigen in vaginal smear.



Treatment

The best treatment or the drug of choice is **metronidazole (Flagyl)** 2 g orally every 8 hours for one week, or 500 mg orally every 12 hours for 7-10 days and refrain from sexual contact. In pregnancy metronidazole is safe in second and third trimesters.

Prophylaxis

Treating the patients that has ***T. vaginalis***, patients sexual partner should be tested for ***T. vaginalis*** when necessary, prevention is same as other sexually transmitted trichomoniasis.



Thank you for listening, any questions?